SIMATS ENGINEERING

ASSESSMENT TOOL 1

Course Name: Computer Networks

Course Code: CSA0706

Course Outcome Covered: CO5: Design network topologies — star, mesh, bus, and hybrid — using networking devices, transmission media, and cabling standards (BL2, BL5)

Assessment: Annexure A — Constraint-Based Problem Solving

Total Marks: 30

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I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Arjun Basava

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Question 1 — Rural Branch Office Connectivity

A multinational company is opening a branch in a remote rural location without high-speed wired internet, needing support for cloud applications, video conferencing, and secure HQ communication, on a limited budget, with uninterrupted operations.

1. Understanding the Problem & Constraints

- Location constraint: no fiber/cable/DSL infrastructure available in the rural area.
- Application constraint: cloud apps (SaaS, ERP) and video conferencing need consistent bandwidth and low latency/jitter, not just raw speed.
- Security constraint: traffic to HQ must be encrypted and authenticated over a public/shared medium.
- Business constraint: uninterrupted operation — a single point of failure is unacceptable.
- Budget constraint: cannot justify enterprise wired trenching or leased lines end-to-end.
- Growth constraint: solution must scale as headcount/traffic grows without a full redesign.

2. Problem Decomposition

- Sub-problem A — Last-mile connectivity: how does the branch reach the internet at all?
- Sub-problem B — Redundancy: what happens if the primary link fails?
- Sub-problem C — LAN and device layer: how do users/devices connect inside the branch?
- Sub-problem D — Secure transport to HQ: how is confidentiality/integrity assured over the WAN?
- Sub-problem E — Cost and scalability: how is spend controlled today while allowing growth?

3. Recommended Solution

A. Primary WAN link: A business-grade satellite broadband service (e.g., a GEO/LEO satellite ISP) or, if within range, a fixed wireless/4G-LTE or 5G fixed-wireless-access (FWA) service. LEO satellite (e.g., low-orbit broadband) is preferred over legacy GEO satellite because it offers lower latency (~25–50 ms vs 600+ ms), which materially improves video conferencing quality.

B. Backup WAN link: A 4G/5G cellular router as a secondary link, configured for automatic failover (SD-WAN or a dual-WAN router). This directly satisfies the 'uninterrupted operations' constraint: if satellite/FWA drops, cellular takes over within seconds.

C. Core networking devices: A dual-WAN capable router/firewall appliance at the edge, an unmanaged or lightly managed switch for wired drops, and a Wi-Fi access point for wireless clients. A small UPS protects the router/switch/AP during power interruptions common in rural areas.

D. Security mechanisms: Site-to-site IPsec VPN (or a lightweight SD-WAN overlay) between the branch router and HQ, so all traffic to headquarters is encrypted end-to-end. A stateful firewall with default-deny inbound rules, WPA3 on the wireless LAN, and multi-factor authentication (MFA) for any cloud application logins add layered protection.

E. Bandwidth management: QoS (Quality of Service) rules on the router prioritize real-time video-conferencing traffic (e.g., DSCP tagging for VoIP/video) over bulk downloads, so a single large file transfer cannot degrade a live call — this addresses the bandwidth-limitation constraint without needing more raw bandwidth.

F. Scalability path: Because the design is built on a VPN/SD-WAN overlay rather than a fixed leased line, adding bandwidth later is a simple ISP plan upgrade, and additional branch sites can be added to the same SD-WAN fabric without re-architecting HQ's network.

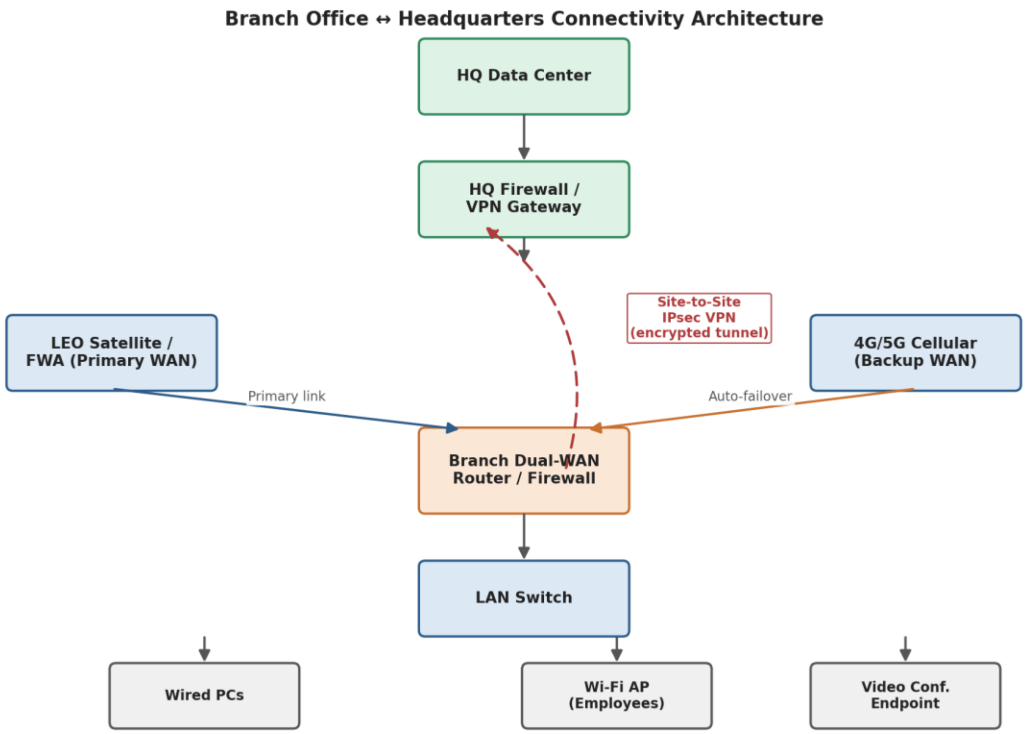

Figure 1.1 — Branch office dual-WAN network architecture with VPN tunnel to HQ

4. Device & Link Specification

Component	Recommended Choice	Purpose
Primary WAN	LEO satellite / 5G FWA modem	Main internet uplink for cloud apps and video conferencing
Backup WAN	4G/5G cellular USB/router modem	Automatic failover if primary link fails
Edge device	Dual-WAN SD-WAN router/firewall	Load balancing, failover, VPN termination, stateful firewall
LAN switch	8–24 port Gigabit unmanaged/smart switch	Wired connectivity for desks and printers
Wireless AP	Wi-Fi 6 access point (WPA3)	Wireless access for laptops, phones, conferencing devices
Power backup	Line-interactive UPS (600–1000VA)	Keeps router/switch/AP running through short outages

5. WAN Selection Decision Flow

The flowchart below summarizes the decision logic used to choose the primary and backup WAN links for the branch, based on what infrastructure is actually available at the rural site.

Figure 1.2 — WAN selection decision flowchart

6. Implementation Roadmap

- Week 1: Site survey — confirm satellite line-of-sight and cellular signal strength at the branch.
- Week 2: Procure and install dual-WAN router, switch, AP, and UPS; activate primary and backup WAN subscriptions.
- Week 3: Configure failover rules, QoS policies, and site-to-site VPN to HQ; test failover by disconnecting the primary link.
- Week 4: Onboard users, enable MFA on cloud applications, and run a live video-conference load test before go-live.

7. Justification Against Constraints

- Bandwidth: LEO satellite/FWA plus QoS keeps latency-sensitive video conferencing usable even on a modest plan.
- Reliability: dual-WAN failover (primary + cellular backup) removes the single point of failure.
- Cost: avoids expensive fiber trenching in a remote area; satellite/cellular hardware and subscription costs are predictable and lower than a dedicated leased line.
- Security: IPsec VPN + firewall + WPA3 + MFA protects data in transit and at the network edge without expensive dedicated security appliances.
- Scalability: SD-WAN/VPN overlay and modular hardware allow bandwidth or additional sites to be added incrementally.

Question 2 — Campus-Wide Wireless Network for 3,000+ Students

A university needs Wi-Fi across classrooms, labs, libraries, and hostels for 3,000+ students, with a limited number of access points (APs), minimal interference, and secure access.

1. Understanding the Problem & Constraints

- Scale constraint: 3,000+ simultaneous users across multiple building types with very different device densities.
- Interference constraint: many APs close together on the same channel cause co-channel interference and slow everyone down.
- Budget constraint: only a limited number of APs can be purchased/deployed — placement must be efficient, not exhaustive.
- Security constraint: only authorized students/staff should get network access, and their traffic must be protected.
- Performance constraint: coverage must translate into usable throughput, not just signal presence.

2. Problem Decomposition

- Sub-problem A — Capacity planning: how many concurrent users per AP, and how many APs are actually needed per zone?
- Sub-problem B — AP placement: where do APs go to maximize coverage per device while minimizing overlap?
- Sub-problem C — Frequency/channel planning: which bands and channels reduce interference?
- Sub-problem D — Authentication and access control: how is each user verified and isolated from unauthorized access?
- Sub-problem E — Cost-performance trade-off: how to hit acceptable coverage/performance within the AP budget?

3. Recommended Solution

A. Capacity-based design (not just coverage-based): Instead of spacing APs purely for signal coverage, size deployment by expected concurrent devices per zone: high-density areas (lecture halls, libraries) get more closely spaced APs (roughly 30–40 devices per AP target), while lower-density areas (hostel corridors, staff rooms) get wider spacing. This uses the limited AP budget where it matters most.

B. AP placement: Ceiling-mounted APs in classrooms and libraries (one per 2–3 average classrooms depending on capacity, or one per large hall), corridor-mounted APs in hostels to serve rooms on both sides, and dedicated APs in labs and auditoriums as high-usage zones. A site survey (heat-map tool) validates actual RF coverage before final placement rather than relying on floor-plan guesses alone.

C. Frequency band and channel selection: Enable both 5 GHz (and 6 GHz where supported by client devices/Wi-Fi 6E) and 2.4 GHz, but bias client load toward 5 GHz/6 GHz using band-steering, since these bands have more non-overlapping channels (e.g., 36, 40, 44, 48… on 5 GHz vs only 1, 6, 11 on 2.4 GHz) and shorter range, which reduces co-channel interference between adjacent APs. Neighboring APs are assigned non-overlapping channels and transmit power is tuned down so cells don't bleed into each other.

D. Authentication methods: WPA2/WPA3-Enterprise with 802.1X, backed by a RADIUS server integrated with the university's existing directory (e.g., LDAP/Active Directory), so each student logs in with their own university credentials — this is enterprise-grade security compared to a shared pre-shared key. A separate, rate-limited, captive-portal guest SSID isolates visitors from the academic/student VLAN.

E. Network segmentation: Separate VLANs/SSIDs for students, staff, labs, and guests, so a compromised guest device cannot reach internal academic systems, and traffic can be prioritized differently per VLAN.

F. Controller-based management: A wireless LAN controller (WLC) or cloud-managed AP system centrally manages channel/power settings, load balancing across APs, and rogue-AP detection — essential at this scale rather than configuring each AP manually.

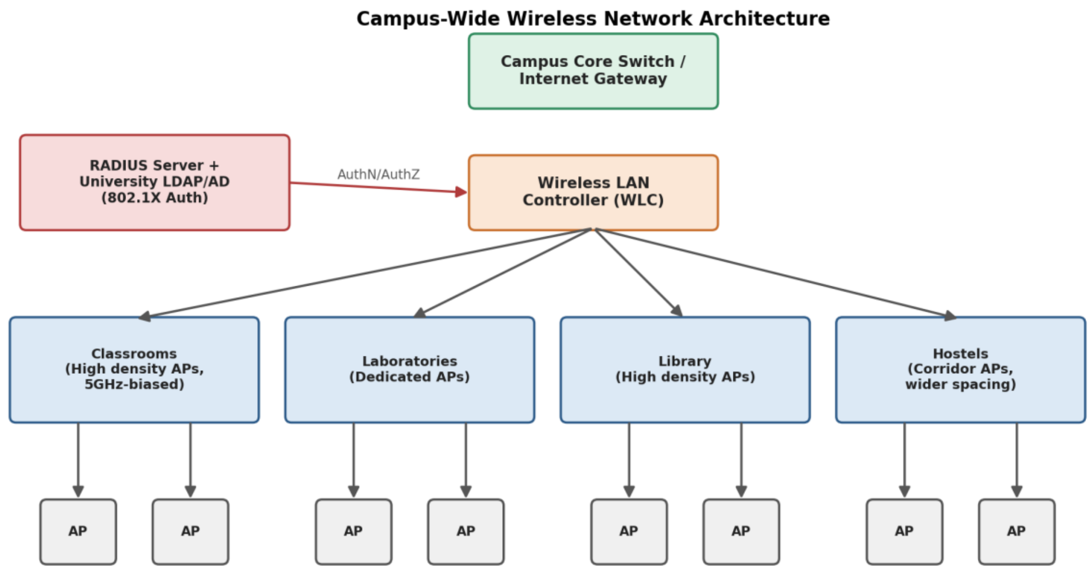

Figure 2.1 — Campus-wide wireless network architecture

4. Zone-Wise AP Allocation Plan

Zone	Expected Concurrent Users	AP Density Strategy	Primary Band
Classrooms	High (30–60 per room)	1 AP per 1–2 rooms, ceiling-mounted	5 GHz / 6 GHz
Laboratories	Medium–High (fixed seats)	1 dedicated AP per lab	5 GHz
Library	High (open study areas)	Overlapping cells for large floor area	5 GHz / 6 GHz
Hostels	Medium (per-room usage)	Corridor APs serving rooms both sides	2.4 GHz + 5 GHz
Auditoriums/Halls	Very high (event-based)	High-density AP cluster, temporary boost	5 GHz / 6 GHz

5. AP Placement & Channel Planning Flow

The flowchart below shows the decision process used to size and tune AP deployment per zone, balancing the limited AP budget against user density and interference.

Figure 2.2 — AP placement and channel planning flowchart

6. Security & Access Segmentation

SSID / VLAN	User Group	Authentication	Access Scope
Campus-Student	Students	WPA3-Enterprise / 802.1X via RADIUS	Internet + academic portals, LMS
Campus-Staff	Faculty & staff	WPA3-Enterprise / 802.1X via RADIUS	Internet + staff systems, restricted admin tools
Campus-Lab	Lab workstations	802.1X + device certificates	Lab servers, internet (filtered)
Campus-Guest	Visitors	Captive portal, rate-limited	Internet only, isolated from internal VLANs

7. Justification Against Constraints

- Coverage: capacity-first placement plus a validated site survey ensures classrooms, labs, libraries, and hostels are all served without wasting APs on low-need areas.
- Performance: band-steering to 5 GHz/6 GHz and controller-based load balancing keep throughput usable even with thousands of concurrent users.
- Interference: non-overlapping channel planning and tuned transmit power directly reduce co-channel and adjacent-channel interference despite dense AP deployment.
- Security: 802.1X/RADIUS authentication with per-user credentials and VLAN segmentation is far stronger than a shared password, at no extra hardware cost.
- Cost: a capacity-driven (not blanket) AP count, combined with centralized management, delivers acceptable performance within a limited AP budget.

Question 3 — Secure Network Architecture for a Financial Institution

A financial institution must upgrade its internal network for stronger cybersecurity while employees access sensitive customer data and online banking, complying with strict security policies, without hurting performance or raising costs significantly.

1. Understanding the Problem & Constraints

- Regulatory constraint: strict compliance requirements (e.g., data protection, financial-sector security standards) govern how sensitive data is stored, accessed, and audited.
- Security constraint: internal and external threats must be detected and blocked without disrupting legitimate banking operations.
- Performance constraint: security controls (encryption, inspection) must not introduce unacceptable latency for time-sensitive banking transactions.
- Cost constraint: the redesign must reuse existing infrastructure where possible rather than a costly rip-and-replace.
- Access constraint: different employee roles need different levels of access to sensitive systems (least privilege).

2. Problem Decomposition

- Sub-problem A — Network segmentation: how is the network divided so a breach in one zone doesn't reach sensitive systems?
- Sub-problem B — Perimeter and internal defense: where are firewalls and intrusion detection/prevention placed?
- Sub-problem C — Authentication and access control: how is identity verified and privilege limited per role?
- Sub-problem D — Monitoring and compliance: how are events logged and audited to satisfy regulators?
- Sub-problem E — Performance/cost balance: how to secure the network without over-engineering or over-spending?

3. Recommended Solution

A. Network segmentation: Divide the network into security zones using VLANs and internal firewalls: a DMZ for internet-facing services (online banking web/app servers), an internal zone for general employee workstations, a highly restricted zone for core banking databases and transaction-processing systems, and a management zone for network administration. This follows a defense-in-depth model — even if the internal employee VLAN is compromised, the core banking zone remains isolated behind its own firewall.

B. Firewall deployment: A perimeter next-generation firewall (NGFW) at the internet edge inspects and filters all inbound/outbound traffic, with a second internal firewall (or firewall zone rules) between the general employee network and the restricted core-banking zone. This two-tier deployment stops external attacks at the edge and limits lateral movement internally, without needing a separate physical firewall per department.

C. Authentication methods: Multi-factor authentication (MFA) for all access to customer data and online banking backend systems, combined with role-based access control (RBAC) so a teller, loan officer, and system administrator each see only what their role requires (principle of least privilege). Single sign-on (SSO) via a central identity provider reduces password fatigue while keeping a single point of strong policy enforcement.

D. Intrusion detection/prevention: An Intrusion Detection and Prevention System (IDS/IPS) is deployed inline at the perimeter and at the boundary of the core-banking zone, using signature-based detection for known threats and anomaly-based detection for unusual internal behavior (e.g., a workstation suddenly querying the customer database at high volume) — catching both external attacks and insider misuse.

E. Encryption and monitoring: TLS encrypts all online-banking traffic in transit; sensitive data at rest in the core-banking database is encrypted. A centralized SIEM (Security Information and Event Management) system aggregates firewall, IDS/IPS, and server logs for real-time alerting and the audit trails regulators require.

F. Performance and cost management: Using existing switches reconfigured into VLANs (rather than new physical LANs per zone) keeps cost low; placing the NGFW/IDS only at zone boundaries (not inline on every single link) limits inspection overhead to where it matters, keeping latency acceptable for transaction processing.

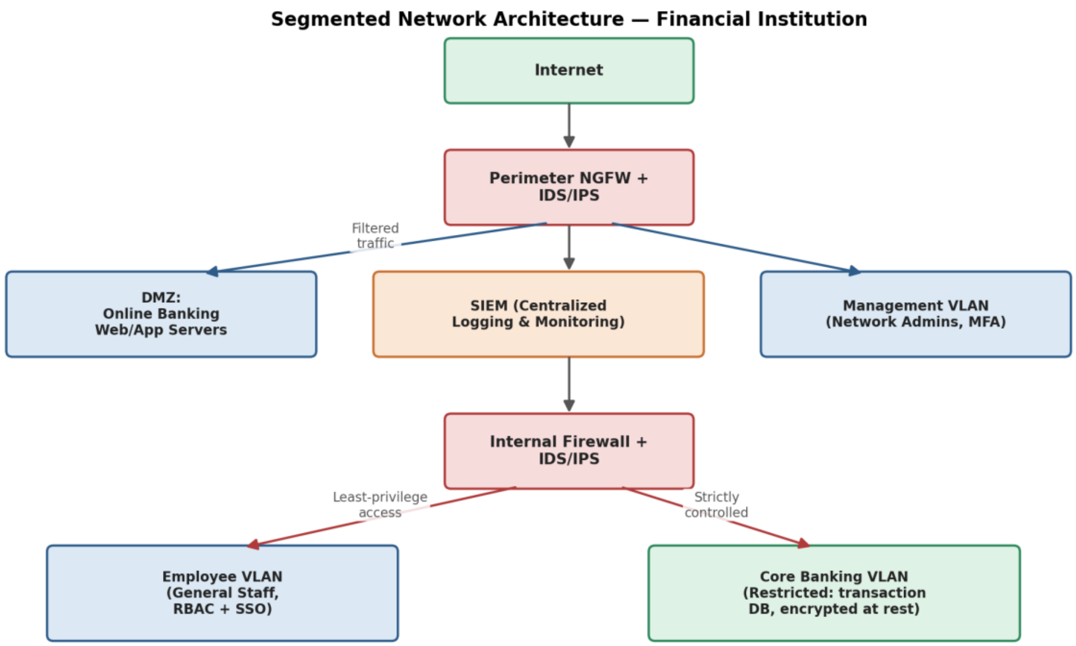

Figure 3.1 — Segmented network architecture for the financial institution

4. Zone / VLAN Access Matrix

Zone	Contains	Inbound Access	Outbound Access
DMZ	Online banking web/app servers	Internet (via NGFW, filtered)	App-tier only, no direct internal access
Employee VLAN	General staff workstations	SSO/MFA authenticated users	Internet (filtered), approved internal apps
Core Banking VLAN	Transaction DB, core banking apps	App-tier + authorized roles only	None to internet; logged to SIEM
Management VLAN	Network admin, monitoring tools	Admins with MFA + VPN	Device management interfaces only

5. Employee Access Control Flow

The flowchart below traces how an employee's request for access is authenticated, authorized by role, and routed through the appropriate security controls before reaching sensitive systems.

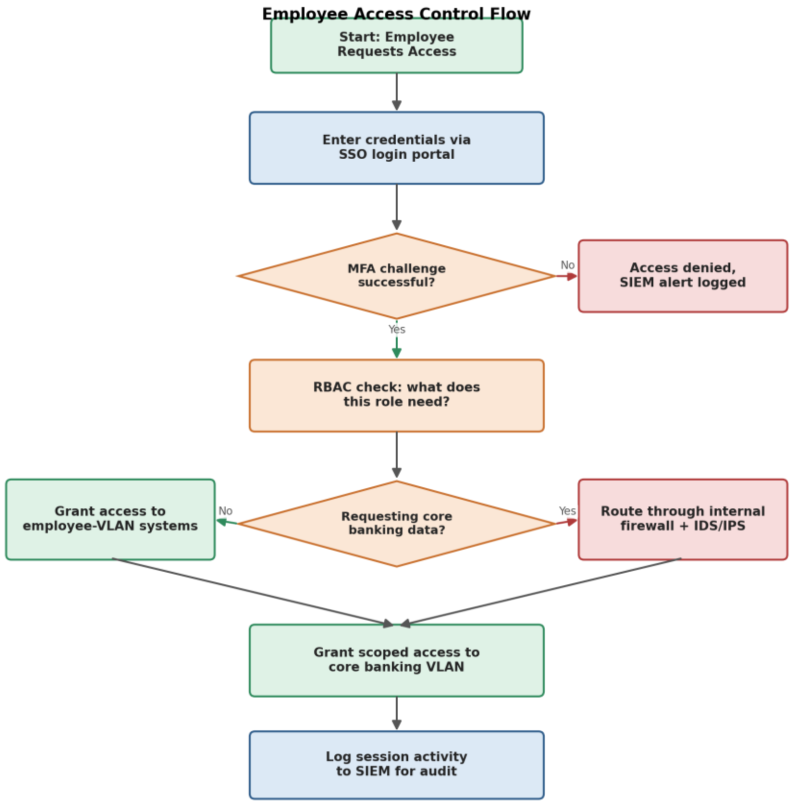

Figure 3.2 — Employee access control and authentication flow

6. Compliance & Monitoring Controls

- Centralized SIEM correlates firewall, IDS/IPS, and application logs to produce the audit trail regulators require.
- Periodic access reviews validate that RBAC assignments still match each employee's current role (no privilege creep).
- Encrypted backups of the core banking database support recovery without weakening the segmentation model.
- Scheduled vulnerability scans and penetration tests validate that zone boundaries hold up against real attack techniques.

7. Justification Against Constraints

- Security: zone segmentation + two-tier firewalls + IDS/IPS provide layered defense against both external attackers and insider threats.
- Performance: inspection is concentrated at zone boundaries rather than every internal link, so day-to-day transaction traffic isn't bottlenecked.
- Compliance: MFA, RBAC, encryption, and centralized SIEM logging directly support the audit trails and access controls that financial regulations require.
- Budget: segmentation is achieved largely through VLAN reconfiguration of existing switches plus firewalls at zone boundaries, avoiding a full physical network rebuild.

Summary: Creativity & Innovation Applied

- Q1 uses LEO satellite + cellular dual-WAN failover rather than a single fixed link, directly targeting the 'uninterrupted operations' requirement.
- Q2 uses capacity-based (not coverage-based) AP sizing so the limited AP budget is spent where user density is highest.
- Q3 concentrates security inspection at zone boundaries only, balancing strong protection against the performance/cost constraint rather than inspecting all traffic uniformly.